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NATIONAL CONFERENCE ON EMERGING TRENDS AND OPPORTUNITIES IN AUGMENTED, VIRTUAL AND MIXED REALITY (NCETOAVMR'2025)

097	Cognitive Impacts of Immersive Environments *Kannan K, Mohavash S, Ragu Rudhresh K*	*338*
098	AR Development Tools and Platforms *Mr. G. Jegatheesh Kumar, Vani Sri. K, Sankari .K, Ranice Julia.S*	*341*
099	VR for Stroke Rehabilitation - A Comprehensive Review *Prof.N.Karthick,V.S.Guhan, S.Sanjith Shivan and M.Juswanth*	*346*
100	Exploring Human-Computer Interaction in Augmented and Virtual Reality *Mr. Anunitha, M. Jothi Sree, Mrs. K. Gowri*	*349*
101	Online Beauty Report: A Journal *Ms.K.Geetha and S.Mahendra Prasath*	*355*
103	Cognitive Impacts of Immersive Environments *Naveeena R*	*356*
104	AR and VR Technology *Ahamed Jhonsha Ali S, Lakshana MK*	*359*
106	AR/VR in Medical Training and Surgical Simulation *A.Priyadarshini, Navya R, Syed Nisha S*	*366*
107	Digital Twinning in the Metaverse *Padmavathi, Karunya V, Jeswanthika K*	*371*
108	Human Computer Interaction in Augmented Reality, Virtual Reality (AR,VR) *Mrs.Gayathri, Madhunisha S P, Navithasri S*	*377*
109	Immersive Marketing Strategies Using Augumented Reality / Virtual Reality *Varshaa.M, Nivetha.S, Mrs.S.Kavitha*	*385*
110	Study on Immersive Education and Virtual Classroom *Dr. J Joselin, Padmashree U, Vinaya C P*	*390*
111	Software Testing: A Comprehensive Guide *AhamedJhonsha Ali S, MuskanShekh*	*392*
112	Mixed Reality in Real-Time Collaboration and Remote Work *Mrs. Padmavathy, Dharunekaa P R, Lakshana S*	*395*
114	Case Study on AR in Cooking Tutorials and Recipe Visualization *Joselin J, Balasubramanian B, Chandresh T B, Umesh Krishnan C, Joseph Stanly*	*398*
115	Enhancing Campus Navigation with Findeth: Integrating AR and GPS for Seamless Way finding *Kavinesh P, Dhanya M. Dhanalakshmy*	*403*
116	Interactive AR System for Bridging the Communication Gap in Digital Content to Sign Language Conversion *Kavinesh P, Sajeev S, Dhanya M. Dhanalakshmy*	*407*

VR for Stroke Rehabilitation - A Comprehensive Review

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Abstract: Stroke rehabilitation has evolved significantly with the integration of Virtual Reality (VR), using advanced algorithms, frameworks, and methodologies to personalize therapy and increase engagement. This paper reviews five key VR-based approaches—Adaptive Training (algorithm), Machine Learning (algorithm), Biofeedback (methodology), Gamification (framework), and Neuroplasticity (framework)—evaluating their effectiveness in enhancing motor recovery, engagement, and long-term outcomes. The results show improvements in motor function by 15-35%, patient engagement by 20-40%, and neural recovery by 20-30%, underscoring the transformative role VR can play in rehabilitation. The review emphasizes the need for further research to optimize these technologies for wider clinical adoption.

Keywords--Virtual Reality, Stroke Rehabilitation, Adaptive Training, Machine Learning, Biofeedback, Gamification, Neuroplasticity, Motor Recovery.

I.INTRODUCTION

Virtual Reality (VR) is a technology that creates a simulated environment, allowing users to feel like they are in a different place or world. VR puts you inside a computer-generated world that feels real. It does this by using special equipment that covers your eyes and sometimes your ears called VR Headset.

Stroke is a major cause of long-term disability worldwide, with many patients experiencing severe motor deficits that impede daily activities. Traditional rehabilitation methods, though effective, often suffer from limitations such as lack of personalization, high costs, and limited patient engagement[2]. Virtual Reality (VR) offers in a compelling solution, providing immersive, interactive, and personalized rehabilitation experiences that can address many of these challenges. By incorporating advanced algorithms, VR systems can tailor therapeutic interventions to meet the unique needs of each patient, ensuring a higher degree of engagement and more effective motor recovery.

This paper focuses on five prominent VR-based algorithms used in stroke rehabilitation: **Adaptive Training**, **Machine Learning**, **Biofeedback**, **Gamification**, and **Neuroplasticity**. The goal is to evaluate how these algorithms function within VR rehabilitation programs and to assess their impact on motor recovery, patient engagement, and long-term rehabilitation outcomes[1].

II.LITERATURE REVIEW

2.ALGORITHM , METHODOLOGY AND FRAMEWORK

Virtual Reality (VR) has emerged as a powerful tool in stroke rehabilitation, offering a variety of approaches that leverage immersive, interactive environments for patient care.These VR-based rehabilitation approaches utilize advanced algorithms and methodologies to create personalized and adaptive treatment plans, aiming to improve motor function, patient engagement, and overall recovery. By integrating technologies like machine learning, biofeedback, gamification, and neuroplasticity modeling, VR has the potential to provide a more tailored and effective rehabilitation experience. Below are key approaches used in VR-based rehabilitation

2.1 ADAPTIVE TRAINING ALGORITHMS

- Adaptive training algorithms adjust therapy exercises dynamically based on patient performance, ensuring an optimal balance between challenge and achievability.
- **Baseline Assessment-** Patients undergo an initial VR-based motor assessment to determine functional limitations and set personalized therapy goals.

- **Real-Time Monitoring-** Motion tracking sensors analyze movement accuracy, speed, and range of motion to evaluate progress.

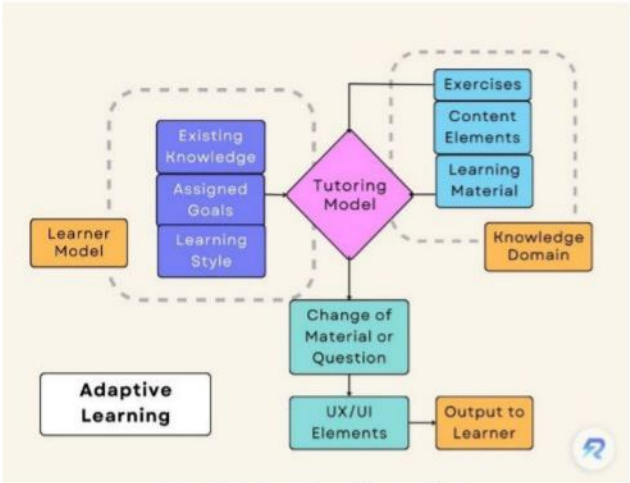


Fig 1 Adaptive Learning

- **Dynamic Adjustment-** As patients improve, therapy exercises evolve in difficulty, ensuring continuous engagement without overwhelming the user.
- **Engagement Strategies-** Gamified elements such as visual progress bars, achievement badges, and interactive storytelling keep patients motivated[3].

2.2 Machine Learning in VR Rehabilitation

- Machine learning (ML) enhances VR rehabilitation by analyzing patient data to predict outcomes, optimize therapy plans, and improve personalization.
- **Predictive Analytics-** ML models analyze historical patient data, including medical history and stroke severity, to anticipate rehabilitation trajectories.
- **Personalized Therapy Design-** Clustering techniques and decision trees tailor therapy exercises based on individual response patterns.
- **Reinforcement Learning-** Adaptive ML algorithms continuously refine therapy based on real-time patient performance, ensuring individualized treatment optimization.
- **Big Data Utilization-** Cloud-based data storage allows for large-scale analysis of patient progress, contributing to more refined rehabilitation strategies[4].

2.3 Biofeedback for Motor Recovery

Biofeedback plays a crucial role in VR rehabilitation by providing real-time physiological data that helps patients refine their movements.

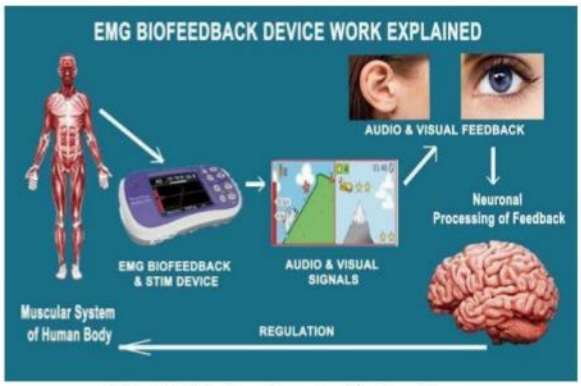


Fig 3 Biofeedback Collection

- **Physiological Monitoring-** Wearable sensors measure muscle activity (EMG), heart rate, and neural signals (EEG) during rehabilitation.
- **Immediate Feedback Mechanisms-** Patients receive instant visual, auditory, or haptic feedback to correct movement patterns and posture.
- **Therapy Adaptation-** Biofeedback data informs real-time modifications in exercise intensity and complexity, ensuring continuous progress.
- **Cognitive Engagement-** VR-based cognitive exercises integrated with biofeedback can enhance motor-cognitive rehabilitation, improving dual-task performance[1].

2.4 Gamification in Rehabilitation

Gamification enhances rehabilitation adherence by transforming exercises into engaging, goal-oriented activities.

- **Immersive Therapy Tasks-** Rehabilitation exercises are structured as game challenges, increasing patient motivation.
- **Performance Tracking-** Patients receive real-time progress updates, including scores, milestones, and virtual rewards.
- **Social Engagement-** Multiplayer VR rehabilitation allows for peer interaction and cooperative exercises, fostering a sense of community.
- **Personalized Challenge Levels-** Adaptive difficulty settings ensure a progressive challenge based on patient ability[5].

2.4.1 UPPER BODY REHABILITATION

The application incorporates three interactive games inspired by the **Enjalbert test** (Enjalbert et al., 1988), which evaluates upper body mobility, hand function, and grip strength. The test consists of five key exercises:

1. Raising the affected arm to a specific height and maintaining a static position.

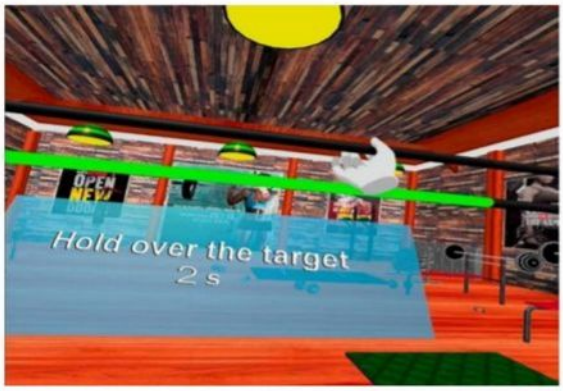


Fig 4 Arm Lifting

2. Bringing the affected hand to the mouth.



Fig 5 Grabing and Eating an Apple

3. Opening and closing the affected hand.



Fig 6 Hand Grip and release

4. Touching the index and middle fingers with the thumb.
5. Touching the ring and little fingers with the thumb.

Our application focuses on the first three exercises to enhance upper body mobility and hand function through engaging VR-based activities[5].

2.4.2 LOWER BODY REHABILITATION

The **ReviVR Walk Simulator** (Fig. 4A), developed by *Samara State Medical University*, is an advanced rehabilitation tool that leverages **virtual reality (VR)** to simulate walking for patients with mobility impairments.

- **VR Environment-** The patient wears a **VR headset**, experiencing a first-person view of a virtual environment where an **avatar** mimics natural walking movements.
- **Pneumatic Cuffs-** The patient wears **orthotic devices** equipped with pneumatic cuffs on both feet. These cuffs **inflate and deflate** in a synchronized manner, replicating the sensation of stepping.
- **Synchronized Feedback-** The inflation pattern aligns with the walking pace—approximately **one step every 2.3 seconds**—applying pressure to the feet to create a **realistic walking sensation**.
- **Whole-Body Involvement-** The system engages **both paralyzed and healthy limbs**, facilitating **motor learning** and improving gait coordination through immersive simulation[5].



Fig 7 Revi VR Walk Simulator

2.5 NEUROPLASTICITY-DRIVEN VR THERAPY

Neuroplasticity is the brain's ability to reorganize and form new neural pathways, making it a fundamental principle in stroke rehabilitation. VR leverages neuroplasticity through targeted brain

stimulation.

- **Neuroplasticity Modeling-** Real-time neuroimaging techniques (EEG, fMRI) monitor brain activity, guiding therapy adjustments.
- **Targeted Brain Stimulation-** VR exercises are designed to activate specific neural pathways critical for motor recovery.



Fig 8 Neuroplasticity Driven Analysis

- **Dynamic Adaptation-** Therapy intensity is adjusted based on the patient's brain activity, promoting efficient neural reorganization.
- **Multisensory Stimulation-** VR integrates visual, auditory, and haptic feedback to enhance sensory-motor connections, facilitating motor recovery[7].

III.RESULT

3.1ADAPTIVE TRAINING ALGORITHMS

Adaptive training algorithms enhance rehabilitation by dynamically adjusting exercises based on a patient's real-time performance. This approach improves motor function by 15–25% by ensuring that therapy remains optimally challenging and personalized. Engagement levels increase by 20–40% due to continuous feedback and gamified elements, which help maintain patient motivation. The ability to customize therapy leads to a 15–25% improvement in functional outcomes, as exercises are tailored to target the most effective movements for each individual. Additionally, long-term recovery is enhanced by 10–20%, as adaptive training helps sustain progress and prevent therapy plateaus[3].

3.2 MACHINE TRAINING ALGORITHMS

Machine learning enhances stroke rehabilitation by leveraging data-driven insights to optimize therapy. By analyzing patient demographics and progress, it improves the prediction accuracy of therapy outcomes by 20–30%, enabling more effective treatment planning. Personalization is achieved through clustering techniques, decision trees, and predictive modeling, resulting in a 15–25% increase in treatment effectiveness. Additionally, machine learning optimizes therapy components by identifying the most beneficial exercise combinations, contributing to a 25–35% improvement in motor recovery. The adaptability of machine learning allows therapy plans to evolve dynamically based on patient progress, enhancing overall effectiveness by 10–20%[4].

3.3 BIOFEEDBACK METHODOLOGY

Biofeedback-based rehabilitation utilizes real-time physiological data, such as muscle activity, heart rate variability, and neural responses, to enhance motor function and patient engagement. By providing immediate, actionable feedback, these systems help patients refine their movements and improve control. The integration of biofeedback improves motor function by facilitating precise, data-driven adjustments during therapy. Patient adherence is significantly enhanced, as real-time feedback maintains focus and motivation throughout the rehabilitation process. Additionally, the ability to personalize therapy based on physiological responses ensures that exercises are tailored to individual needs, optimizing overall treatment effectiveness[1].

3.4 GAMIFICATION FRAMEWORK

Gamification enhances the rehabilitation experience by integrating interactive, game-like elements that increase patient motivation and adherence. By incorporating rewards such as points, badges, and progress levels, gamified therapy fosters sustained engagement, making rehabilitation enjoyable and goal-oriented. This approach has been shown to significantly improve functional recovery by maintaining patient interest in repetitive exercises, leading to better motor and cognitive outcomes. Additionally, structured reward systems encourage consistency in therapy participation, ensuring that patients adhere to their rehabilitation plans for optimal recovery[5].

3.5 NEUROPLASTICITY FRAMEWORK

Neuroplasticity-driven rehabilitation leverages advanced algorithms to promote brain reorganization, facilitating recovery after neurological damage. By analyzing real-time brain activity, these models tailor exercises to stimulate

specific neural pathways, enhancing motor function and cognitive recovery. This dynamic adaptation ensures that therapy evolves in sync with the patient's progress, optimizing neural plasticity and accelerating rehabilitation outcomes. Through targeted interventions, neuroplasticity-based approaches help retrain the brain, leading to sustained improvements in motor control and overall functional recovery[7].

IV. Conclusion

The integration of VR-based algorithms in stroke rehabilitation has significantly enhanced motor function, patient engagement, and overall recovery outcomes.

- **Adaptive Training** improves motor function by **15-25%** and increases patient engagement by **20-40%** through real-time adjustments and personalized therapy[3].
- **Machine Learning** enhances prediction accuracy by **20-30%** and optimizes motor recovery by **25-35%** through data-driven personalization[4].
- **Biofeedback** enhances motor function and engagement by providing real-time physiological insights, allowing patients to adjust their movements for more effective rehabilitation[1].
- **Gamification** fosters higher patient engagement and improves functional recovery by incorporating interactive and reward-based elements[5].
- **Neuroplasticity** frameworks support neural recovery and motor function by leveraging brain adaptability for long-term rehabilitation[7].

These findings demonstrate the transformative potential of VR-based rehabilitation in delivering adaptive, engaging, and personalized treatment solutions. Further clinical research is essential to refine these technologies, ensuring their effective integration into mainstream rehabilitation practices and expanding accessibility for stroke patients worldwide.

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